

# **X-RAY PROPERTIES OF CATAclysmic VARIABLE STARS**

**Darren Stuart Baskill, M.Phys.**

**Supervisors:**

Peter Wheatley

Julian Osborne

Mike Watson

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X-ray Astronomy Group  
Department of Physics and Astronomy  
University of Leicester

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# Chapter 1

## Introduction

”[looking upwards] There’s a new star in heaven tonight... A new freckle on the nose  
of the giant pixie.”

Baldrick,  
Blackadder III, BBC TV

### 1.1 Discovery of Cataclysmic Variables

Stars have been known to vary in brightness since ancient times, with naked eye supernovae acting as rare and dramatic reminders, yet coherent catalogues of varying stars did not appear until the telescope had become commonplace. During the middle of the 19<sup>th</sup> century, there was great activity in the search for planets after the discovery of Neptune in 1846. On the night of the 15<sup>th</sup> of December, 1855, J.R.Hind was searching for minor planets around the ecliptic when he noticed a previously unrecorded ninth magnitude star in the constellation Gemini (Hind 1856) - he had discovered a cataclysmic variable star that had suddenly brightened from its usually much fainter state (hence the reason for it being unrecorded). Unlike other variable stars known at that time, either Mira pulsating stars or the Algol binary systems, U Geminorum (as it became known) was unusually blue. U Gem was observed to brighten again three months later and became a popular object to observe, yielding an almost complete optical light curve through to the present day. It was found to brighten dramatically (an *outburst*) within a day, for 5-20 days, every 2-8 months. It was not until 1896 that another similarly behaving star, SS Cyg, was discovered by Miss L. D. Wells at the Harvard Observatory. Today, over 800 cataclysmic variables have been discovered, and they are expected to be commonplace throughout the galaxy - approximately  $10^6$  are thought to exist within our galaxy alone.

## 1.2 The Optical Observations

Ever since U Gem and SS Cyg were discovered, the long term monitoring of the brightness of variable stars has been left to the amateur astronomy community. All the optical data included in this thesis originates from amateur astronomers worldwide, who measure the magnitudes of stars by comparing with nearby stars of known (and stable) magnitude. Groups such as the BAA-VSS (the British Astronomical Association Variable Star Section) formed in 1891, and the AAVSO (the American Association of Variable Star Observers), founded in 1911, have proved invaluable in collecting visual observations of these variable stars. They maintain large databases of the observations made by amateur astronomers (over 300,000 observations are collected per year by the AAVSO alone) which are then made accessible for professional research.

Only recently have long-term professional optical monitoring of star brightness's begun. Projects such as ROTSE (Marshall *et al.* 1997), ASAS (Pojmanski 1997) and TASS (Droege *et al.* 1997), look for optical variations in a wide variety of objects from dwarf novae, novae and supernovae through to the optical counter-parts of gamma-ray bursts, asteroids and extra-solar planets. Although successful, they are still very much in the early stages of development.

## 1.3 Light-curve Classifications

Variable stars are primarily classified by how their visual brightness varies with time. The defining characteristic of the dwarf nova class of cataclysmic variable stars are the outbursts that reoccur on time-scales from  $\approx 10$  days to many years (over 400 dwarf novae are known, including both U Gem and SS Cyg). During these outbursts, which occur at unpredictable intervals and last between 2-20 days, the system can increase in brightness by typically 2-5 magnitudes - a luminosity increase of up to 100 fold. Dwarf novae are divided into three subclasses, named after the first star discovered of that class; the U Gem, SU UMa and Z Cam classes. Dwarf novae are classified into these three groups depending on the morphology of their optical light curve.

- *U Geminorum stars*: U Gem is the prototype of a class of dwarf novae that undergo outbursts with typical amplitudes of 2-6 magnitudes (see figures 1.1 & 1.2). The outbursts rise within 1-2 days, with the shape and duration varying between successive outbursts in what appears to be a bimodal distribution in many (Cook 1987, Ak, Ozkan & Mattei 2002), although all the outbursts have the same decay time. The time between successive outbursts are anything between ten and thousands of days. Throughout this thesis, I use the U Gem classification to denote those stars that are neither SU UMa or Z Cam type dwarf novae (some authors use the term U Gem as a synonym of dwarf novae, using the SS Cyg type to differentiate from those that are not SU UMa or Z Cam types), and the class contains some 60 members.
- *SU Ursae Majoris stars*: In addition to the regular outbursts of dwarf novae which last 1-2 days, the 90 known SU UMa stars also display occasional *superoutbursts* - outbursts which last 10-20 days and are up to 2 magnitudes brighter than the “normal” outbursts (see figure 1.4 and section 1.6.1). During superoutbursts, small 0.2-0.3 magnitude oscillations (*superhumps*) are observed (see section 1.6.1). The superoutbursts usually occur once every 3-4 normal outbursts, although they are notable exceptions such as the WZ Sge systems.



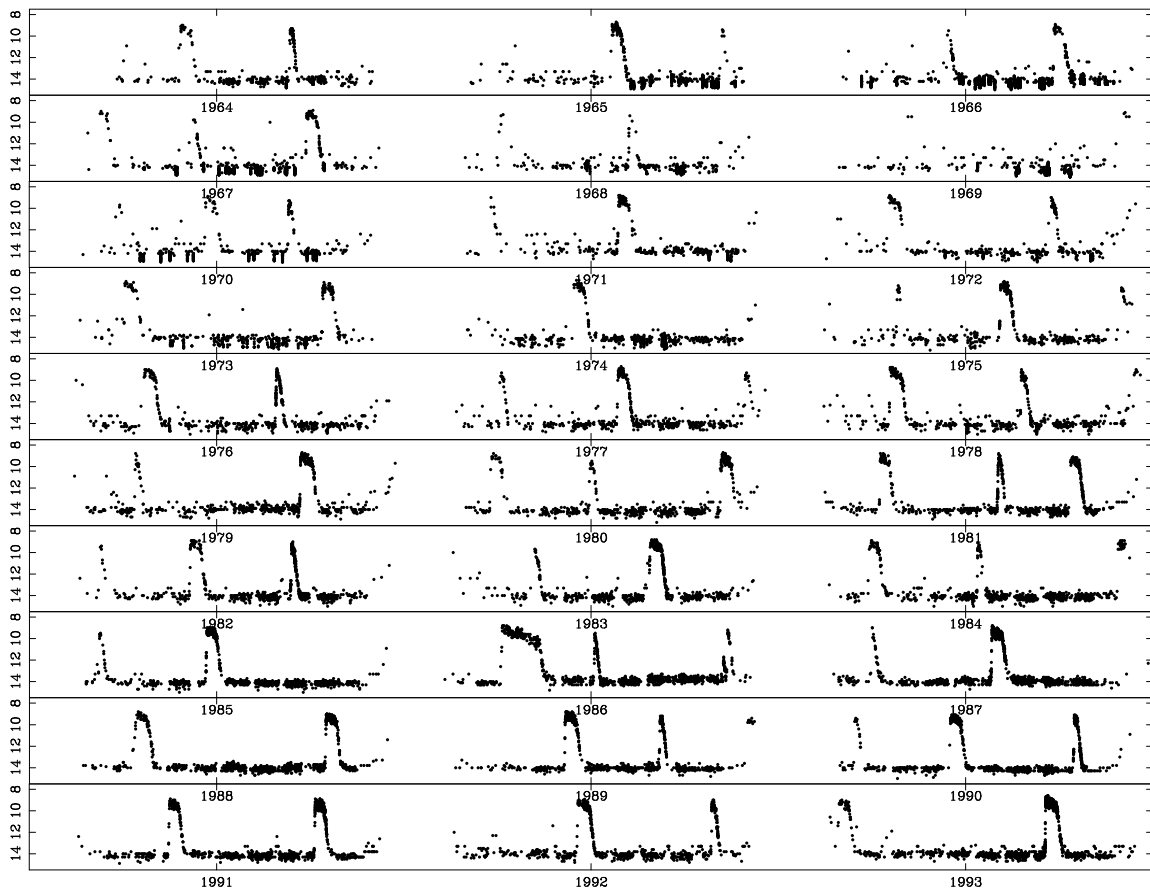


Figure 1.1: Optical light-curve of U Gem from AAVSO observations taken between 1963 and 1999. Since Gemini is one of the zodiacal constellations, U Gem cannot be observed for approximately two months around conjunction with the Sun.

- *Z Camelopardalis stars*: As these dwarf novae optically fade from an outburst, Z Cam stars can occasionally reach a *standstill*, where the brightness of the star remains almost constant somewhere between the outburst and quiescent values (see figure 1.3). These standstills can last from tens of days to years. Eventually the brightness of the star falls to the quiescent value, and the outburst cycles begin once more. Almost 40 Z Cam systems have been discovered to date.

## 1.4 The Geometry of Non-Magnetic Cataclysmic Variables

The nature of cataclysmic variables remained unknown until the later half of the twentieth century. Elvey & Babcock (1943) measured radial velocities of SS Cyg and noted that the “...microphotometer tracings resemble lines which are produced through high axial rotation... If these lines are produced through rotation of the star it would appear that the rotational velocities are much higher than would be allowed from stability considerations.”. Spectroscopic observations of the nova AE Aquarii (Joy 1954) and photometric observations of DQ Herculis (Walker 1956), led to the discovery that novae were short period binary systems (DQ Her was observed to be an eclipsing binary). Kraft (1962) studied five U Gem type stars and spectroscopically showed that they too were binary systems with orbital periods

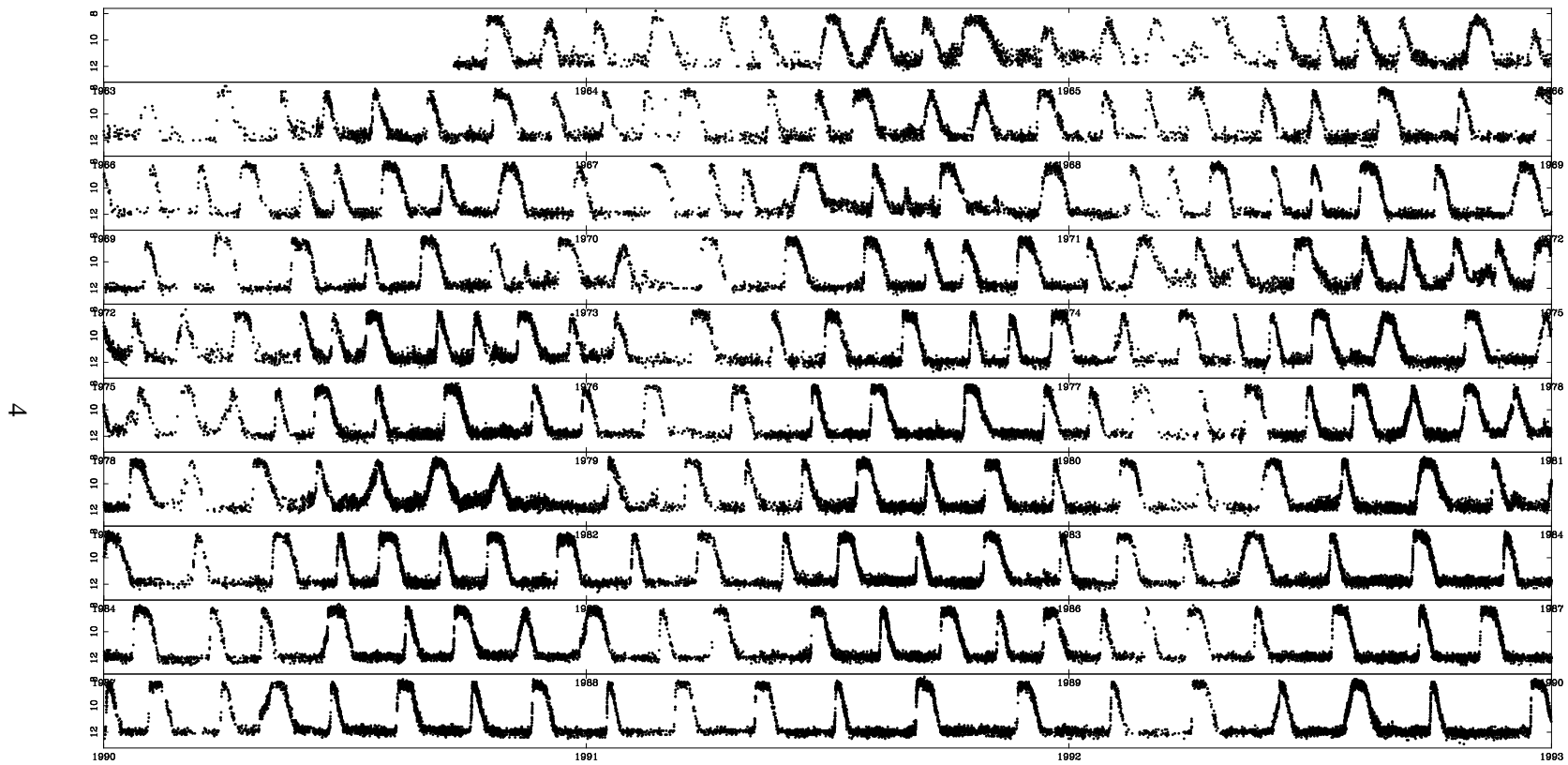


Figure 1.2: Optical light-curve of SS Cyg from AAVSO observations taken between 1963 and 1999 (the data is available as an AAVSO monograph).

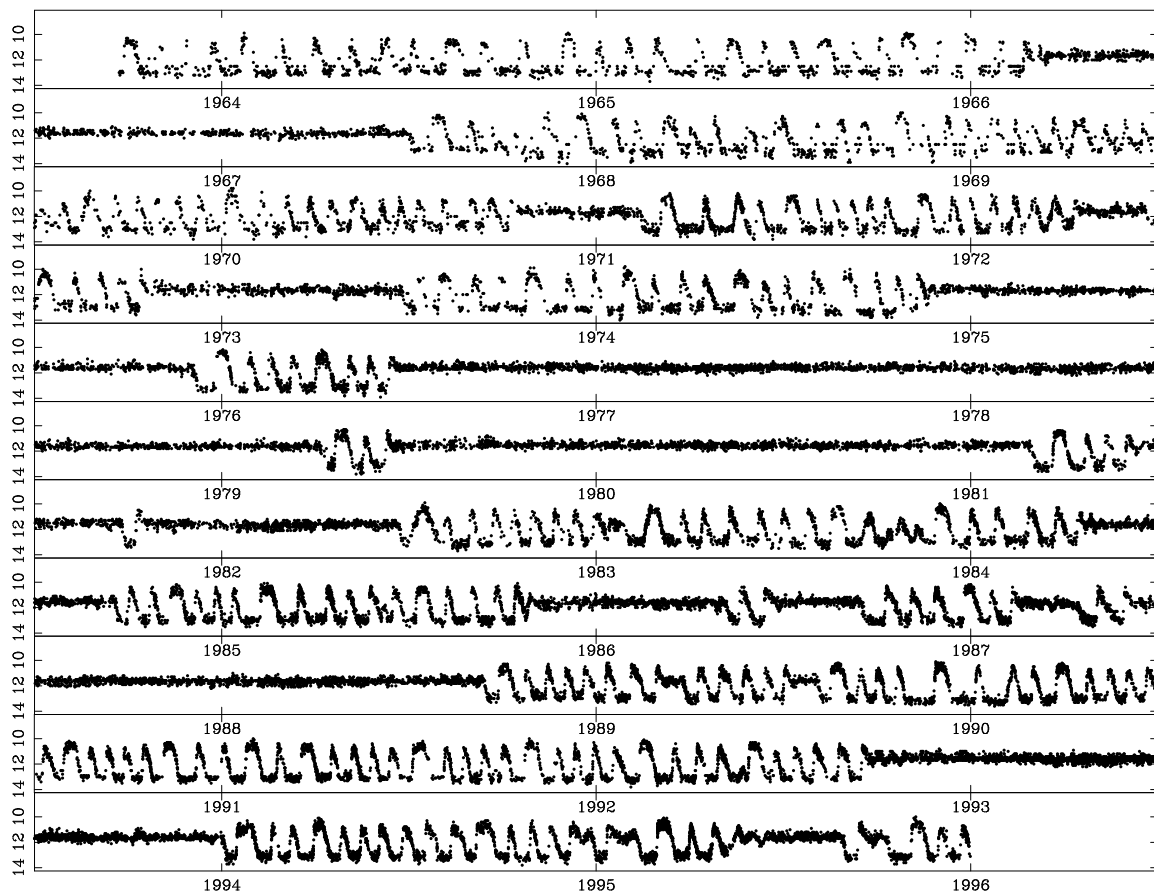


Figure 1.3: Optical light-curve of Z Cam from AAVSO observations taken between 1963 and 1999 (the data is available as an AAVSO monograph).

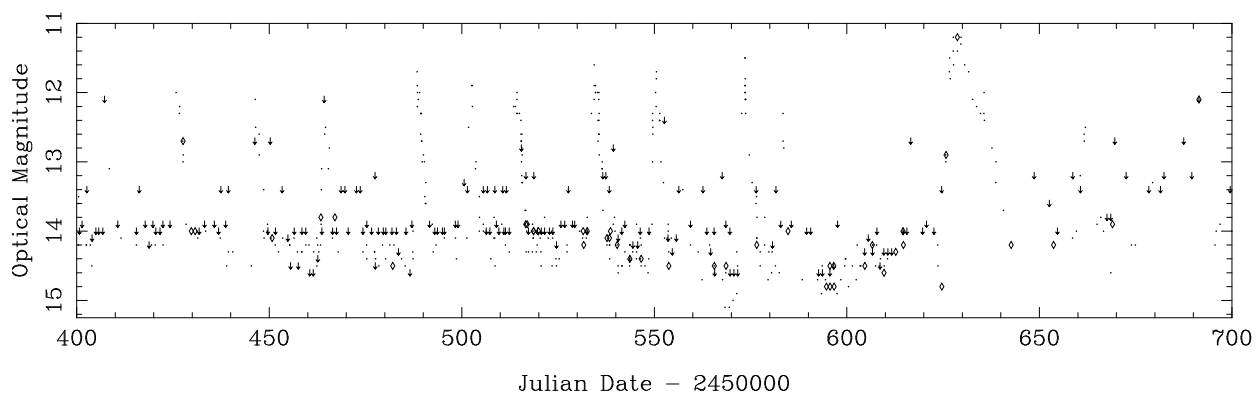


Figure 1.4: Optical light-curve of SU UMa from AAVSO observations made between November 1996 and August 1997. A superoutburst occurred on JD 2450625.

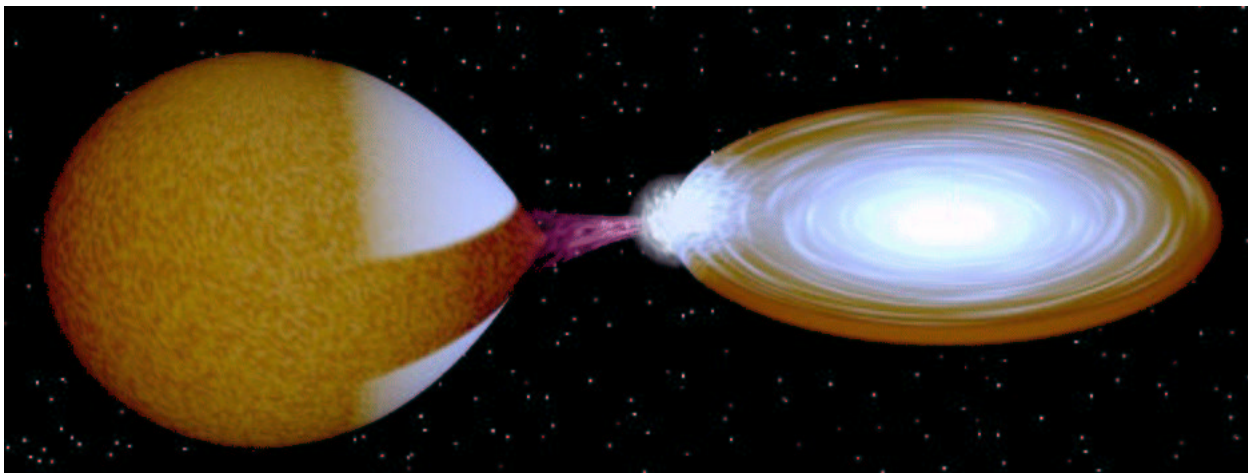


Figure 1.5: Artists impression of a non-magnetic cataclysmic variable (image courtesy of Rob Hynes).

of  $<9$  hours. In the same paper, Kraft (1962) suggested that “.. the red stars overflow their lobes of the inner Lagrangian surface; the ejected material forms, in part, a ring, or disk, surrounding the blue star.”, which he suggested was a white dwarf. This conclusion was based on the under-luminosity of the secondary star and through comparisons with the W UMa type stars (a contact binary of two similar type stars within a common envelope) which had been recently shown to overflow their inner Lagrangian surface by Kitamura (1959). Further spectroscopic investigation of SS Cyg led to the same conclusion when doubling was detected of the K-line of Ca II, “..probably because of a rotating ring or shell of material surrounding the hot star.” (Walker & Chincarini 1968). High speed photometry of Z Cha by Warner (1974) finally proved through eclipse observations that the outburst was centred on the primary, suggesting that a disk did exist around the white dwarf. This confirmed the geometric picture of Warner & Nather (1971) who had observed a bright spot which they suggested to be on the edge of a disk surrounding the primary.

Through these (and other) observations, a consistent picture of the geometry of non-magnetic cataclysmic variable systems<sup>†</sup> has evolved (see figure 1.5). The secondary star fills its Roche lobe (the surface that defines the region that is gravitationally bound to the star, and over which the strength of the gravitational field is constant) and material spills out towards the white dwarf<sup>‡</sup> through the point where the gravitational force from each star balance (the inner Lagrangian point). This material is prevented from directly falling onto the white dwarf surface due to the orbital momentum of the in-falling material and instead falls into orbit around the white dwarf. Viscosity effects then causes the stream to spread out as the orbiting material loses energy due to this internal friction, slowly spiralling inwards and (to conserve angular momentum) outwards until an *accretion disk* forms around the white dwarf. As material passes through the disk, gravitational potential energy is converted into thermal and kinetic energy, and the disk is heated to higher temperatures nearer to the white dwarf. Finally, the material collides with the surface of the white dwarf, at a rate of  $\approx 10^{12} - 10^{15} \text{ kg s}^{-1}$ .

<sup>†</sup>The term “non-magnetic” refers to the white dwarf primary star. All cataclysmic variables are believed to evolve due to the magnetic field of the secondary star (see section 1.7), and systems with magnetic white dwarf primaries are discussed in section 1.8

<sup>‡</sup>For more about white dwarfs in cataclysmic variables, see the excellent review by Sion (1999).

## 1.5 The origin of the Dwarf Nova Outbursts

It can be shown that the luminosity of the disk is directly proportional to the mass transfer rate through the disk. Hence, the outbursts of dwarf novae appear to be caused by a sudden avalanche of material falling through the disk. Half of the energy available to a particle falling through the inner Lagrangian point is expected to be released in the disk, resulting in the other half being released in a region known as the *boundary layer* between the inner disk and the white dwarf surface (see section 1.10). To prove this, we first equate the gravitational force to the centrifugal force acting on a mass  $m$  of orbiting gas in a stable orbit within the accretion disk:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

The gravitational constant  $G = 6.671 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ ,  $M$  is the mass of the white dwarf,  $v$  is the velocity of the particle around the accretion disk and  $r$  is the radial distance from the centre of the white dwarf. Now consider the total energy of this mass of gas, which is the sum of the kinetic energy and gravitational potential energy (through substituting of the force equation):

$$E = \frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2r}$$

This result is a consequence of the Virial theorem, which states that the gravitational potential energy is twice the negative of the kinetic energy of a system of particles in equilibrium. Now consider the energy of an annular ring of gas at radius  $r$  and with a thickness of  $dr$ . The total mass of the ring is equal to the mass accreted through the disk in a time  $t$ ; that is,  $m = \dot{M}t$ . Hence:

$$dE = \frac{dE}{dr} dr = \frac{d}{dr} \left( -\frac{GMm}{2r} \right) dr = \left( \frac{GM\dot{M}t}{2r^2} \right) dr$$

Thus, the luminosity of the ring is (assuming that energy escapes at the point where it is generated):

$$dL_{ring} = \frac{dE}{t} = \frac{GM\dot{M}}{2r^2} dr$$

i.e. the luminosity of the ring is directly proportional to the mass transfer rate through the disk. Integrating for the luminosity of each ring between the inner disk radius and infinity gives the total luminosity of the disk:

$$L_{disk} = \int_R^\infty dL_{ring} = \int_R^\infty \frac{GM\dot{M}}{2r^2} dr = \frac{GM\dot{M}}{2R}$$

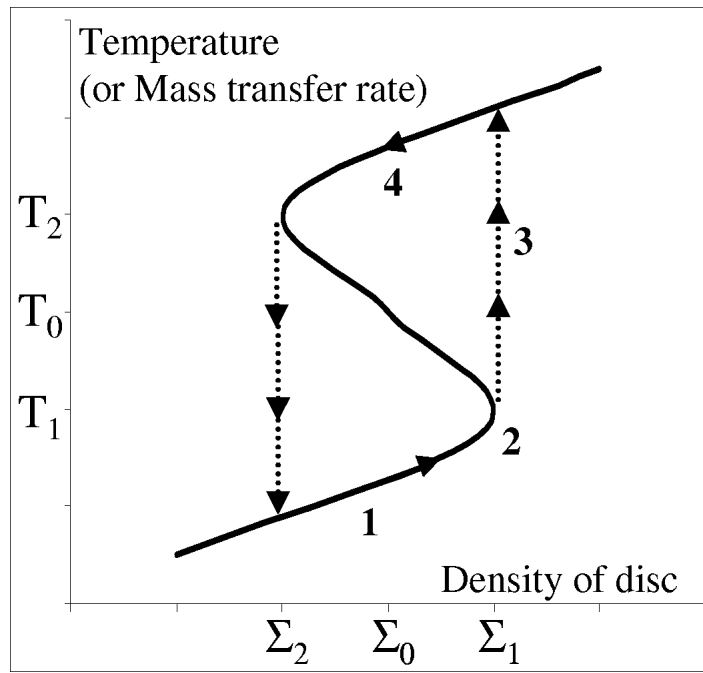


Figure 1.6: The theoretical explanation for the outburst cycles in dwarf novae. The “S-curve” represents the thermal equilibrium of the accretion disk, as a function of density and temperature (see text for further description).

However, for a particle that is simply dropped from a height above the white dwarf without a disk, the energy released is just the initial gravitational potential energy,  $E = GMm/r$ . The resulting accretion luminosity in this case would be  $L_{acc} = GM\dot{M}/R$ . Thus, when an accretion disk is present, half of the available energy is expected to radiate from the disk, as material spirals down towards the white dwarf. The other half of the accretion energy (if the white dwarf is not rapidly rotating) must therefore be released as the material finally reaches the white dwarf surface, as X-rays produced in the boundary layer. However, observations suggest that this may be an over-simplified treatment of the accretion disk, since most non-magnetic cataclysmic variables appear to be under-luminous in the X-ray band when compared with the optical and ultraviolet luminosity (I discuss this in more details in chapter 6, section ??).

The standard model used to describe the outburst cycle observed in dwarf novae is the *disk instability model*. During quiescence, viscosity in the disk (and hence the mass transfer rate) remains low, allowing matter to accumulate in the disk. The outbursts are believed to occur when the viscosity in the disk suddenly increases, causing the mass transfer rate through the disk (and hence luminosity) to increase.

The disk instability model was originally proposed by Osaki (1974), and the subsequent development of the model is covered in the review by Osaki (1996). Referring to figure 1.6, the outburst cycle is believed to behave as follows. Material slowly builds up in the disk, as the mass transfer rate from the secondary star is greater than the mass transfer rate through the inner disk and onto the white dwarf (point 1). This build up of gas in the disk causes the temperature and density of the disk to slowly rise. Once the temperature gets high enough ( $T_1$  and point 2 in figure 1.6), there is a further rapid increase in temperature as the disk transfers to the hot stable branch (point 3), which causes the viscosity in the disk to also increase. This is believed to be caused when the accretion disk temperature is high enough

for hydrogen to be ionised, which increases the opacity of the gas (the photons collide with the free electrons). The high opacity results in inefficient cooling and hence the rapid temperature increase to the stable hot branch. Since both the viscosity and temperature are now high, the mass accretion rate through the disk increases and an avalanche of material drains the disk onto the white dwarf faster than the supply from the red star (step 4). Finally, the disk returns to its initial state, with a low temperature and density, and the cycle repeats.

The middle branch of figure 1.6 corresponds to unstable equilibria - if the local temperature of any point in the accretion disk lies between  $T_1$  and  $T_2$  the disk cannot remain stable. A stable disk only occurs on the upper hot (radiative) branch or the lower cool (convective) branch. For the cool branch, the surface density of the disk cannot exceed  $\Sigma_1$ , and for the disk to be on the hot branch, the surface density of the disk cannot fall below  $\Sigma_2$ . Both  $\Sigma_1$  and  $\Sigma_2$  are functions of the viscosity, the white dwarf mass and radial distance from the white dwarf.

The exact nature of this viscosity that causes the outbursts is currently unknown. The most promising explanation for the source of the viscosity comes from the work of Balbus & Hawley (1991), who propose that the disk may be dynamically unstable to axisymmetric disturbances in the presence of a weak magnetic field. The degree of coupling between the magnetic field and a partially ionised gas is a very sensitive function of temperature near the transition point of good and poor coupling. This causes a runaway - a temperature increase lowers the resistivity, allowing more turbulences and more heating; a temperature decrease generally increases the resistivity, damping the turbulence and reducing the heat source still further (Balbus 2002).

In the simple disk instability model that I have outlined, the rate of mass transfer from the secondary is assumed to be constant. This may not be the case (see the review of Lasota 2001, who discusses the problems of the disk instability models in detail). For the disk instability model (DIM) to reproduce the light curves of dwarf novae accurately, irradiation of the secondary star and the accretion disk, secondary star induced tidal instabilities in the disk, and heating at the stream-disk impact site all need to be taken into account. As Lasota (2002) concludes, "Modern' DIM's discs are truncated and/or irradiated, tidal-torque and stream-impact heating of the outer disc is taken into account and the rate of mass supply from the (irradiated) secondary varies like share prices on the Stock Exchange."

One common addition to the disk instability model, used to explain the prolonged super-outbursts observed in the SU UMa types systems (section 1.3), are tidal instabilities. The accretion disk expands during outburst, increasing its size with each successive outburst. The accretion disk expands until after several outbursts it reaches a critical disk radius, where a disk in a low-mass secondary system becomes eccentric due to the repeated tidal pull from the secondary star. A resonance instability occurs where the orbital period of an annulus of the accretion disk is one third of the binary orbital period, producing a two spiral armed density wave (Osaki 1996). Once the outer disk reaches the critical radius, enhanced tidal torques act on the disk during the next outburst to create the much bigger superoutburst. This behaviour has been successfully simulated by Truss, Murray & Wynn (2001).

## 1.6 The Physical Origins of the Classifications

I have already discussed the classification of dwarf novae in terms of their optical light curves in section 1.3. Here, I discuss the sub-classifications of the dwarf novae, novae and novae-likes in terms of the disk instability model (for a review, see Osaki 1996). In figure 1.7, I show how the different non-magnetic cataclysmic variable types (as defined by the behaviour of their optical light curves) can all be explained by the disk (and tidal) instability model, using differing mass accretion rates and orbital periods.

### 1.6.1 The Dwarf Novae

#### U Geminorium (U Gem) Stars

The U Gem type dwarf novae (as already previously mentioned in section 1.3; see also figures 1.1 and 1.2) usually have a quiescent (low state) accretion disk, and outbursts occur between every ten to hundreds of days. In terms of the disk instability model, such outbursts are expected to originate in long orbital period ( $> 3$  hours), low accretion rate ( $\dot{M} < 10^{16} \text{ g s}^{-1}$ ) systems. The outbursts occur due to the disk instability as described in the previous section, without the intervention of the tidal instability; the two stars in U Gem systems have a large enough separation that the accretion disks do not expand enough for the tidal forces of the secondary to become important.

#### Z Camleopardalis (Z Cam) Stars

The Z Cam stars undergo periods of standstill, where the optical luminosity stabilises between the usual outburst and quiescent values for up to many years (see section 1.3 and figure 1.3). The disk-instability model predicts such an effect to occur in those systems where the mass transfer rate from the secondary increases, which suppresses the outbursts. During standstill, the accretion rate from the secondary star is believed to be higher than during the outburst cycles, and so the temperature of the disk will be just above  $T_2$  and the surface density just above  $\Sigma_2$  in figure 1.6 - the standstill occurs on the upper hot branch. If the mass accretion rate from the secondary falls, the disk returns to the cold branch (Z Cam stars always return to the quiescent state at the end of standstill). This variation may be an effect of irradiation of the secondary star, producing higher mass accretion rate into the disk during standstill (Meyer & Meyer-Hofmeister 1983).

#### SU Ursae Majoris (SU UMa) Stars

SU UMa type dwarf novae, all of which have periods of less than 2 hours with a low mass ratio (secondary mass over primary mass), undergo superoutbursts - an outburst which lasts 5-10 times longer than the usual outbursts, are brighter by approximately a magnitude, and occur typically once every 3-4 normal outbursts. The superoutbursts are believed to be caused by a tidal instability. Superhumps (small, 0.2-0.3 magnitude amplitude, optical oscillations on top of the superoutburst light curve) are



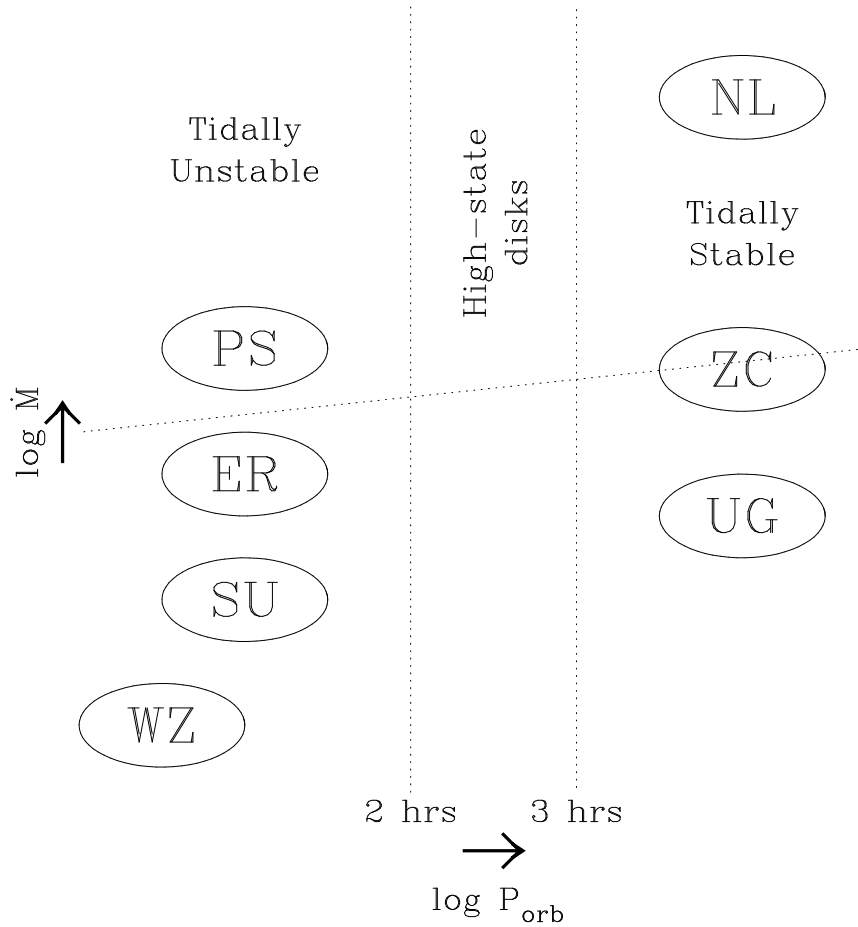


Figure 1.7: Figure taken from Osaki (1996), showing how the disk instability model can be used to explain the different behaviour of non-magnetic cataclysmic variable sub-classifications. The variable types (described in detail in section 1.6) are: WZ Sge (WZ), SU UMa (SU), ER UMa (ER), permanent superhumpers (PS), U Gem (UG), Z Cam (ZC) and the nova-likes (NL). The outer accretion disks of systems below the 2-3 hour period gap are truncated through tidal resonance effects with the secondary star (the tidally unstable systems, see section 1.6.1), whereas systems above the period gap are tidally stable (the timescale for the disks to reach the larger tidal truncation radius are so much longer that superoutburst have never be observed). The accretion disks of systems with mass transfer rates above the critical mass accretion rate (dotted line) are in a permanent high state, whereas below the line systems are also observed to have a low quiescent state.

observed in all SU UMa type (and superoutbursting) stars, with a period of a few percent longer than the orbital period, which are observed only during superoutburst. The superhumps occur at the beat period between the precessional period of the elliptical disk and the binary orbital period. The elliptical accretion disk slowly precesses with a period  $P_P$ , the secondary star orbits with a period  $P_{orb}$ , and the period between when the secondary star and the accretion disk both present their brightest phases occurs at the superhump period,  $P_{sh}$ , and are related thus:  $\frac{1}{P_{sh}} = \frac{1}{P_{orb}} - \frac{1}{P_P}$ .

SU UMa type stars are further divided into two subclasses; ER UMa and WZ Sge types.

- *ER Ursae Majoris (ER UMa) Stars* (Osaki 1996) have an extremely short period of time between each super-outburst (usually between 19-44 days), a high duty cycle (that is, the stars spend 30-50% of the time in a superoutburst state), the normal outbursts occur on short periods between the superoutbursts (usually every 3-4 days) and the superoutbursts are of small amplitude (about 3 magnitudes).
- *WZ Sagittae (WZ Sge) stars* (O'Donogue *et al.* 1991) have an extremely long period between successive outbursts (of order years), long-lasting outbursts (weeks to months), and large amplitude superoutbursts (6-9 magnitudes). WZ Sge stars are also known as TOADs - Tremendous Outburst Amplitude Dwarf-novae (Howell, Szkody & Cannizzo 1995). Normal outbursts in WZ Sge stars are very rare, and have never been observed in WZ Sge itself.

## 1.6.2 Novae & Nova-likes

Novae are cataclysmic variables that are observed to brighten by 8-15 magnitudes, a much more dramatic eruption than the dwarf nova outbursts. When compared with dwarf nova outbursts, nova eruptions are rarely seen (on average, only 3 eruptions are observed in our galaxy per year) and are caused by a very different process. In novae, the primary star accretes hydrogen rich material from the secondary via a disk, and builds up a layer of hydrogen on the surface of the white dwarf primary. The base of this layer is compressed and heated by the gravitational potential of the white dwarf until a critical temperature and pressure is reached, enabling thermonuclear reactions. A runaway process takes place, and the surface of the white dwarf explodes blowing a shell of material away from the primary star<sup>†</sup>. Recurrent novae are those novae where this process (and the emitted shells) have been repeatedly observed. The nova-likes are simply those stars that resemble quiescent novae (or post-novae) spectroscopically and in their light-curves (showing small amplitude fluctuations), although no nova eruption has yet been observed. Novae are believed to have a high accretion rate, and so no accretion disk outburst cycles are expected (or have been observed) in-between the classical novae eruptions since the accretion disk is in permanent outburst (see section 1.5).

In the following sections I describe the classification of the novae and nova-likes, which are based on their optical behaviour in-between the nova eruptions.

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<sup>†</sup>Throughout this thesis, I refer to such events as novae *eruptions*, to differentiate them from the accretion disk *outbursts* observed in dwarf-novae.

### **UX Ursae Majoris (UX UMa) Stars**

The accretion disk in UX Ursae Majoris (UX UMa) stars (originally used as a synonym of nova-likes) appear to be in a permanent high state and show persistent absorption line spectra (Warner 1995). The UX UMa stars may be Z Cam stars in permanent standstill (Meyer & Meyer-Hofmeister 1983). The RW Triangulum (RW Tri) stars which display slightly different spectral properties appear to be UX UMa type stars observed at low inclinations. A sub-class of the RW Tri stars are the SW Sextantis (SW Sex) type stars (Thorstensen, Davis & Ringwald 1991), which display absorption features at certain orbital phase.

### **VY Sculptoris (VY Scl) Stars**

The VY Sculptoris (VY Scl) stars (also called anti-dwarf novae, a term coined by Belserene 1981) vary around a mean magnitude before occasionally fading by several magnitude for periods of days to years (see figure ??), during which they appear spectrally similar to dwarf novae in quiescence. During maximum light, the VY Scl stars behave the same as the UX UMa, RW Tri or SW Sex stars. In terms of the disk instability model (see figure 1.7), these stars lie close to the Z Cam type dwarf novae but with a higher accretion rate.

### **Permanent Superhumpers**

The permanent superhump stars are nova-likes with short orbital periods and a large accretion rate through the disk. They are in permanent outburst and show superhumps in their optical light curves, as seen in the SU UMa type dwarf novae (see 1.6.1).

### **AM Canis Venaticorum (AM CVn) Stars**

AM CVn systems are nova-likes that are helium-rich systems (with no hydrogen lines) with a very short orbital period (e.g. 15-50 minutes). Only 11 such systems have been discovered. In such systems, both the primary and secondary are believed to have evolved into white dwarf stars, and accretion occurs via a disk (see section 1.7).

## **1.7 The Evolution of Cataclysmic Variables**

Cataclysmic variables evolve from detached binary systems that have wide separation of several astronomical units and long orbital periods of order years. However, the two components are of different masses, allowing the more massive member to evolve more rapidly than the other until it swells to become a red giant (see Warner 1995, Patterson 1984 and Charles & Seward 1995 for overviews). The giant (primary) star expands to the point where it engulfs the secondary (normal) star within its tenuous atmosphere, causing the secondary to spiral inwards due to atmospheric drag, causing a rapid

decrease in both the orbital period and separation of the two stars. The common envelope atmosphere is believed to dissipate rapidly, leaving behind the core of the red giant (now a white dwarf star) and the main-sequence secondary in a close binary.

The orbital period of the binary decreases further (although more slowly than during the common envelope stage) via magnetic braking (Eggleton 1976, Weber & Davis 1967). Magnetic braking is the process whereby mass from the stellar wind of the secondary is locked along the magnetic field lines, and angular-momentum is lost to the stellar wind causing the binary orbit to tighten<sup>†</sup>. As the two stars get closer, the secondary star overfills its Roche lobe (the volume that defines the gravitationally bound region of the star), and mass transfer between the two stars begins.

During the early stages of cataclysmic variable evolution, the thermal time-scale ( $t_{th}$ ) of the secondary (the time required for the secondary to adjust so that it remains in thermodynamic equilibrium) is much less than the mass transfer time-scale ( $t_{MT}$ ) of the secondary (the time it takes the secondary to react to mass loss), and so the secondary can remain in thermal equilibrium and on the main sequence.

However, as the secondary continues to loose mass, the thermal time-scale increases (the evolution of the secondary to thermal equilibrium slows), until the system reaches an orbital period of  $\approx 3$  hours where  $t_{th} > t_{MT}$ . The secondary can no longer remain in thermodynamic equilibrium - the secondary cannot keep up with the rate at which it is losing mass. Thus, as the orbital period of the system decreases further, the secondary evolves off the main-sequence due to the effects of mass loss - the mass transfer has forced the secondary out of thermal equilibrium. The star is over sized and therefore over luminous for its new mass, which reduces the core temperature, and the secondary leaves the main sequence.

A histogram of the period of known cataclysmic variables (see figures ?? and ??) shows a lack of cataclysmic variable systems between 2-3 hours. This is known as the period gap problem. Why are no CV's detected in this gap? A common explanation is that the binary becomes detached, and mass transfer between the two stars has ceased.

Once the secondary evolves to become fully convective, the strength of magnetic field is believed to drop rapidly<sup>†</sup> (Spruit & Ritter 1983, Rappaport, Joss & Verbunt 1983). As the angular momentum losses reduce, the evolution of the system slows, and the secondary is allowed to return to thermodynamic equilibrium once more ( $t_{th} < t_{MT}$ ). If the radius of the secondary now falls to thermodynamic equilibrium faster than the Roche lobe shrinks, the system would become detached and mass transfer between the two stars would cease. It is this evolutionary picture that is often used to explain the period gap problem. Note that the period gap actually represents a deficit in numbers, with several cataclysmic variables existing with orbital periods between 2-3 hours; such systems are believed to have been "born" (mass transfer initiated) in the period gap, and do not experience a gap since they never left thermodynamic equilibrium (there was no mass loss forcing the secondary out of equilibrium when the orbit period was above 3 hours).

The rate at which angular momentum is lost from a system increases once again as the CV evolves to

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<sup>†</sup>Tidal torques force the secondary into synchronous lock, where its orbital period is identical to its rotational period (a minimum energy state for the system). Hence, angular momentum cannot be lost to the stellar wind from the rotation of the secondary, only from the binary orbit.

<sup>†</sup>However, recent results from open cluster surveys (Pinsonneault, Andronov & Sills 2002) shows no evidence for a change in observed angular momentum losses at the fully convective boundary.

even lower orbital periods, when angular momentum losses via gravitational radiation are thought to dominate. Gravitational radiation is a prediction from Einstein's general theory of relativity, and occurs when the distribution of mass varies. This changes the surrounding curvature of space and time, and hence extracts both energy and momentum from a system.

As the binary orbit continues to tighten and the orbital period falls below 2 hours, the Roche lobe of the secondary shrinks to the point where mass transfer between the two stars begins once again. The system has evolved through the period gap without mass transfer, and then re-emerges once again as a cataclysmic variable but below the period gap.

The secondary star may eventually evolve into a degenerate brown dwarf star (e.g. Paczynski & Sienkiewicz 1981 & Rappaport, Joss & Webbink 1982), once the secondary mass is low enough. In such a system, mass transfer still occurs between the two stars via an accretion disk, but hydrogen burning in the secondary has ceased as the star relaxes thermally to become fully degenerate. The orbital period of the system reaches a minimum at  $\approx 78$  minutes, and then begins to increase again (the so-called orbital period bounce); this is due to the difference between the mass-radius relations of main-sequence dwarfs and degenerate dwarfs<sup>†</sup> (Rappaport, Joss & Webbink 1982, Faulkner 1971).

The double degenerate AM Canis Venaticorum systems are thought to evolve via a different path, with the secondary in these systems having also evolved through the red giant phase. The white dwarf primary spirals into the expanding red giant secondary, and drives off the envelope (producing a planetary nebula) leaving the red giant's degenerate core orbiting the white dwarf (Nather, Robinson & Stover 1981). At this late stage of CV evolution, so much angular momentum has been lost from the system that the two white dwarfs orbit each other in a very tight orbit, with orbital periods of less than a hour. Mass transfer begins from the degenerate core of the red giant, and an accretion disk forms around the white dwarf primary. AM CVn type stars continue to evolve via gravitational radiation, and mass transfer may continue until the less massive component is complete devoured, resulting in just a single degenerate object - possibly a DB white dwarf star (Nather, Robinson & Stover 1981).

## 1.8 Magnetic Systems

All the cataclysmic variables studied in this thesis are non-magnetic systems; those CVs where the white dwarf has a magnetic field strength less than  $0.1 - 1$  MG. However, I will briefly introduce the magnetic systems here for the sake of completeness. In CV systems where the white dwarf is strongly magnetic ( $B > 10$  MG), no disk can form around the white dwarf primary, and the accretion is magnetically funnelled directly onto the magnetic poles of the white dwarf - such systems are called polars, or AM Her systems (see figure 1.8). Because there is no disk, polars do not exhibit accretion disk outbursts. In systems where the white dwarf primary is mildly magnetic ( $B \approx 1 - 10$  MG), an accretion disk may exist (and hence accretion disk outbursts are sometimes observed) but the inner disk is truncated depending on the strength of the field (see figure 1.9). The material is accreted via accretion columns that reach from the inner disk directly onto the magnetic poles of the white dwarf. These systems are known as intermediate polars, or DQ Her systems.

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<sup>†</sup>It can be shown that the orbital period is solely dependent on the density of the secondary ( $P \propto \rho^{-\frac{1}{2}}$ ), using Kepler's third law and the secondary mass-radius relation of Patterson (1984) (see King 1988). As the secondary loses mass its density increases (the lower density outer layers are accreted away). Once the secondary becomes degenerate, further mass loss *reduces* its density (a lower pressure and hence density is required to degenerately support a less massive white dwarf).

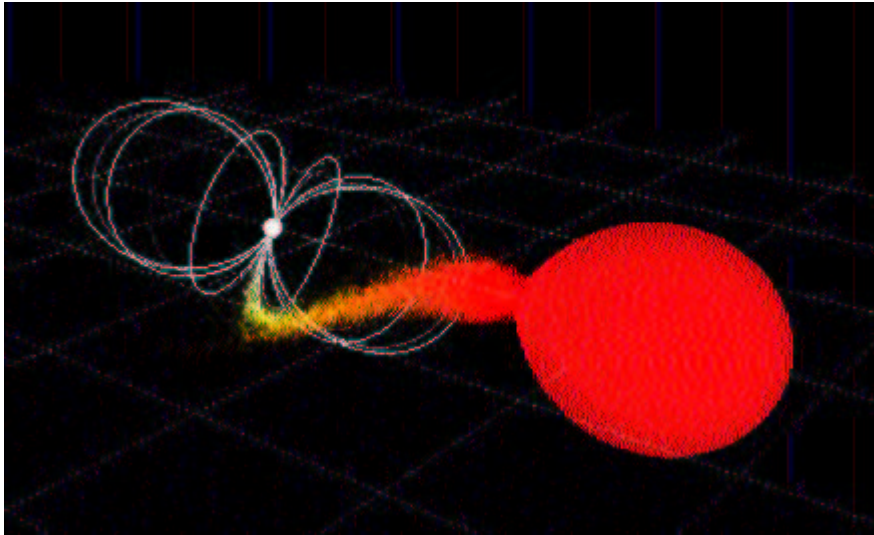


Figure 1.8: The polars (or AM Herculis stars) are cataclysmic variable systems that consist of a strongly magnetic white dwarf, where the accretion is magnetically funnelled directly onto the magnetic poles of the white dwarf, without an accretion disk. This simulation was carried out by Graham Wynn and Richard West, using the UKAFF super-computing facility at the University of Leicester.

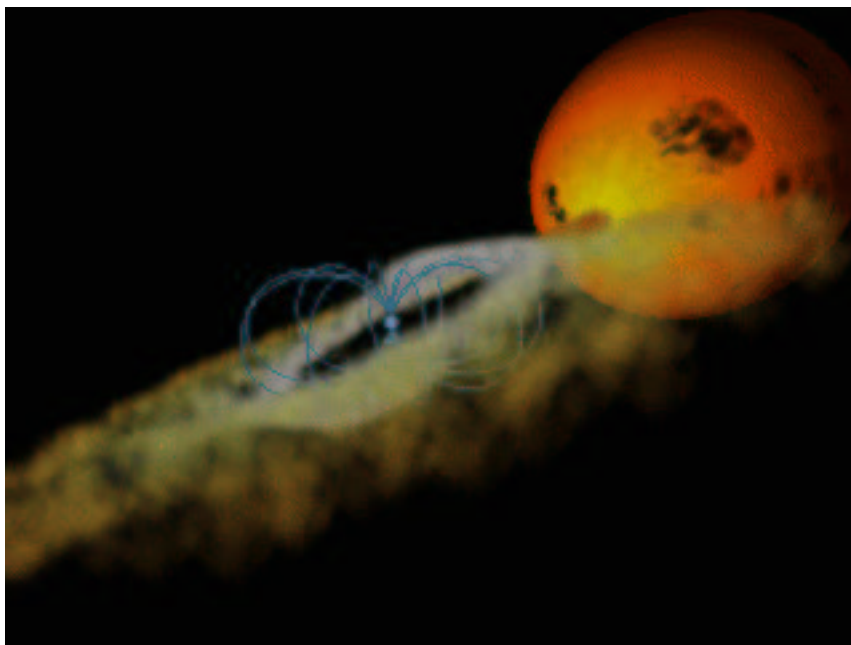


Figure 1.9: The intermediate-polar systems (or DQ Herculis stars) are cataclysmic variable systems that consist of a weakly magnetic white dwarf. An accretion disk exists, but the inner disk is truncated where accretion is magnetically funnelled onto the magnetic poles of the white dwarf (image courtesy of Andrew Beardmore).

## 1.9 Historical Overview of X-ray Astronomy

Making observations of the X-ray emission from dwarf novae (the origin of which is described in the following section 1.10) require space-borne instrumentation since X-rays cannot penetrate the atmosphere of the Earth - just 10cm of air (at one atmosphere pressure) is enough to stop 90% of X-rays at a typical energy of 3 keV (Charles & Seward 1995). V2 rockets captured after the second world war led to development of X-ray observations of the Sun. It was not until 1962 that other sources of X-rays were discovered. It was thought that solar X-rays may reflect off the surface of the Moon, causing the lunar surface to fluoresce, and leading to important insights into the surface composition in advance of men walking on the Moon at the end of that decade. Although the experiment failed to detect X-rays from the Moon, it did detect X-rays from a cosmic source: Scorpius X-1. Optically, Sco X-1 appears as an inconspicuous 13<sup>th</sup> magnitude star.

In 1967-68, two groups surveyed the southern skies with rocket-borne telescopes (as described in Charles & Seward 1995). The group at Lawrence Livermore Laboratory detected the X-ray source Cen X-3. However, when the group at Leicester University (led by Ken Pounds) repeated the experiment, Cen X-3 was not detected. This apparent contradiction was resolved in 1971, when the *Uhuru* satellite was launched and investigated the variability of Cen X-3 in detail.

*Uhuru* detected a 4.8 second periodicity in Cen X-3. This period was observed to vary due to a Doppler shift, proving that the X-ray source was orbiting around an unseen companion every 2.1 days. The X-rays also disappeared for 11 hours every orbital period proving that the system was an eclipsing binary system - the secondary star was eclipsing the X-ray emission (which explained the earlier detections and non-detections). By equating the gravitational and centrifugal forces acting on a particle, it can be shown that the distance of this particle from the centre of mass must be less than  $(GMP^2/4\pi^2)^{1/3}$ . Through estimating the mass of the object, only a rotating neutron star could produce such a rapid period as observed in Cen X-3, since a white dwarf has a radius of  $\approx 10^4$  km.

The first cataclysmic variable to be detected at X-ray wavelengths was the magnetic intermediate-polar EX Hya during an all sky survey using the *Uhuru* satellite (Warner 1972). In 1973, soft X-rays were discovered (using rocket-borne instruments) from the dwarf nova SS Cyg (Rappaport *et al.* 1974), the optically (and X-ray) brightest dwarf novae in the sky.

In the following sections, I discuss some of the more successful X-ray observatories, and in particular their relevance to the study of non-magnetic cataclysmic variables (see also section ?? for a detailed description of previous work).

- *Uhuru* operated between December 1970 and March 1973, and was the first Earth-orbiting mission dedicated entirely to celestial X-ray astronomy. With an energy range of 2 - 20 keV, and only being sensitive to the brightest X-ray sources, it did not detect any X-rays from dwarf novae (although a possible detection of SS Cyg in the *Uhuru* 2-6 keV band was noted after *HEAO-1* detections, Fabbiano *et al.* 1978). However, *Uhuru* did observe other (X-ray brighter and harder) cataclysmic variables, including EX Hya (Warner 1972).
- The British-USA *Ariel 5* made observations between October 1974 and March 1980, making X-ray observations between 0.3-40 keV. The on-board all sky monitor allowed long-term monitoring of many X-ray sources. Although many X-ray binary systems were discovered (containing

neutron stars and black holes), SS Cyg was the only dwarf nova bright enough to be detected (Watson, Sherrington & Jameson 1978).

- The launch of *HEAO-1* resulted in the first systematic survey of soft X-ray emission from cataclysmic variable stars (Cordova, Jensen & Nugent 1981). The energy range of 0.2-10 keV was ideally suited to observations of dwarf nova - if the targets were luminous enough. Through cross-correlation of known cataclysmic variables with X-ray detections, Cordova, Jensen & Nugent (1981) concluded that CVs must be generally low-luminosity X-ray sources, with fluxes below a few  $\times 10^{-11}$  erg cm $^{-2}$  s $^{-1}$  in the 0.1-3 keV band. *HEAO-1* operated between august 1977 and January 1979.
- The *Einstein* observatory (or *HEAO-2*) was the first imaging X-ray observatory, operating between the end of 1978 until spring 1981. With an energy range of 0.2-20 keV, a 100 fold increase in sensitivity, and an angular resolution of a few arcseconds, *Einstein* was a significant improvement over any X-ray observatory that had gone before it. The key importance of this mission in developing the field of X-ray observations of non-magnetic cataclysmic variables was its sensitivity. Cordova & Mason (1984) discovered X-ray emission from 18 out of 31 cataclysmic variables observed, showing that the majority had hard ( $kT > 2$  keV) X-ray spectra and variable X-ray flux on timescale of tens of seconds to hours.
- Launched in May 1983 and operational for 3 years, *EXOSAT* observed non-magnetic cataclysmic variables from an highly eccentric 90-hour orbit, allowing long uninterrupted observations. With an energy range that proved ideal for observing CVs (0.05-20 keV), the principle results of the X-ray observations of dwarf novae are presented in Mukai & Shiokawa (1993). However, the low effective area of the telescopes on-board *EXOSAT* resulted in the majority of dwarf novae being marginal or non-detections. Bremsstrahlung spectral modelling with temperatures between 2-20 keV can adequately describe all the observations of dwarf novae (Mukai & Shiokawa 1993).
- During the first six months, *ROSAT* carried out a soft X-ray (0.1 - 2.5 keV) survey of the entire sky, detecting over 150,000 objects (including comets!). A further 8 years of observations then followed, allowing deep pointed observations of individual targets with an X-ray telescope some 1000 times more sensitive than that on-board *Uhuru*. The principle results of the *ROSAT* observations of non-magnetic cataclysmic variables are presented by Van Teeseling, Beuermann & Verbunt (1996), Verbunt *et al.* (1997) and Richman (1996). The later present detailed spectral fitting and flux estimates of 37 cataclysmic variables, suggesting that X-ray emission in CVs is not well described using single temperature models. Richman (1996) also detect orbital hardness modulations in 17 of the 37 CVs.

The Japanese have a long history of building and operating Earth-orbiting X-ray observatories spanning four decades, which have made a significant contribution to our knowledge of X-ray astronomy. In the following sections, I discuss the Japanese contribution to X-ray astronomy.

- *Hakucho* (Japanese for Swan) was launched in February 1979 and operated for 6 years, covering the energy range of 0.1 - 100 keV. However, *Hakucho* was only sensitive to the brightest X-ray sources.
- *Tenma* (meaning “Pegasus”) was operational between February 1983 and late 1984, covering the 0.1 - 60 keV energy range. Its spectral resolution allowed a distinction to be made between the



reflection iron line (off cold material at 6.4 keV) and the 6.7 keV iron line complex in x-ray bright active galactic nuclei and X-ray binaries, although the sensitivity was not adequate for studies of non-magnetic cataclysmic variables.

- *Ginga* (the Japanese word for galaxy) was launched in 1987, and had a four and a half year lifespan, observing X-rays between 1 - 400 keV. Many observations were made of non-magnetic CVs, including SS Cyg (Yoshida, Inoue & Osaki 1992, Done & Osborne 1997), VW Hyi (Wheatley *et al.* 1996), V426 Oph (Szkody, Kii & Osaki 1990) and LS Peg (Szkody *et al.* 1994).
- In 1994 *ASCA* was launched, with an energy range of 0.6-10.0 keV it was the first X-ray observatory to use CCD detectors (see chapter 3). The name is taken from *Asuka*, an ancient Japanese word meaning flying bird and an era in Japanese history, which was Americanised to the acronym *Advanced Satellite for Cosmology and Astrophysics*. *ASCA* was optimised for X-ray spectroscopy, combining imaging capability with a broad spectral band, good spectral resolution, and a large effective area; thus it complimented *ROSAT* (optimized for X-ray imaging) and *RXTE* (optimised for timing studies).
- On the 10<sup>th</sup> of February, 2000, *Astro-E* (the fifth in this series of Japanese X-ray observatories) was launched. Unfortunately, there was a problem with the first stage of the launch rocket, and *Astro-E* could not obtain the necessary altitude for a proper orbit. A relaunch is currently planned for 2005, with three instruments covering the range 0.4-700 keV; a X-ray micro-calorimeter spectrometer, X-ray imaging spectroscopic CCDs and a hard X-ray detector.

To bring this discussion up to date (as of time of writing in 2002), I will also briefly describe the latest generation of X-ray observatories, and introduce relevant papers. Although, unlike *ASCA*, these observatories have yet to observe a large sample of non-magnetic cataclysmic variables, they are progressing research in the field through providing higher quality observations.

- The *Rossi X-ray Timing Explorer (RXTE)* was launched on 30 December, 1995. *RXTE* specific aim is to monitor the temporal variability of X-ray sources over a large range of timescales (from milliseconds to months), with moderate spectral capabilities between 2-250 keV. Although non-magnetic cataclysmic variables are not bright enough to be detected with the All-Sky Monitor, *RXTE* also performs pointed observations with the Proportional Counter Array. The weakness of *RXTE* is that it is a non-imaging telescope, and the background subtraction is performed via modelling of the background. Although this is not a problem for bright sources, where the source brightness dominates the background, for faint source (especially dwarf novae in outburst) accurate determination of the background level is difficult to achieve. Examples of results from *RXTE* can be seen in figures ??, reffig:suumaxte, and ??. *RXTE* has well studied the dwarf novae SS Cyg (the X-ray brightest dwarf nova; see Wheatley, Mauche & Mattei 2003, and Priedhorsky, Trudolyubov & McGowan 2002), and V1159 Ori (Szkody *et al.* 1999; a superoutburst and eight normal outbursts were observed), as well as the nova V382 Vel (Mukai & Ishida 2001).
- The Italian *BeppoSAX* X-ray observatory was launched in April 1996<sup>†</sup>. The novelty of this mission was the 0.1-300 keV broad spectral range, with a relatively large effective area, medium energy resolution and imaging capabilities in the range of 0.1-10 keV. Papers based on *BeppoSAX* data have been published for the dwarf nova VW Hyi (Hartmann *et al.* 1999, who show

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<sup>†</sup> *BeppoSAX* ceased operating in April 2002, 6 years to the day after launch.

significant differences between outburst X-ray light-curves), and the novae V382 Vel (Orio *et al.* 2002, Orio *et al.* 2001) and U Sco (Kahabka *et al.* 1999).

- The high special resolution ( $< 1$  arcsecond) of NASA's *Chandra* observatory allows images of X-ray sources to be seen with a clarity never before achieved, which is also coupled with a high spectral resolution. Its large elliptical orbit allows long uninterrupted observations of X-ray sources. Although *Chandra* has only been operational for 2.5 years, papers have already been published with results to interest astronomers in the field of non-magnetic cataclysmic variables. Papers have been published based upon *Chandra* observations of the dwarf novae U Gem (Szkody *et al.* 2002), SS Cyg (Mauche 2002), and WZ Sge during the unexpected 2001 outburst (Steehls 2001), as well as the recurrent nova CI Aql (Greiner & Di Stefano 2002).
- The *XMM-Newton* observatory was launched in December 1999 into a large 48 hour elliptical orbit allowing long uninterrupted observations. The *XMM-Newton* mirrors have an unprecedented large effective area, allowing *XMM-Newton* to be sensitive to even the faintest X-ray sources. The on-board EPIC cameras cover the spectral range of 0.1-15 keV, and the on-board optical monitor allows simultaneous optical coverage - both of which are for the study of non-magnetic cataclysmic variables. Although *XMM-Newton* had observed 6 non-magnetic cataclysmic variables by the end of 2001, only the results of the OY Car observation has been published. In that quiescent observation, Ramsay *et al.* (2001) observe a 2240 second X-ray modulation which they attribute to the spin period of the white dwarf, and analysis by Wheatley & West (2002) show the X-rays originating from high latitude regions on the white dwarf (which may be due to either the presence of a weak magnetic field, or absorption of the equatorial emission).

## 1.10 X-ray Emission from Non-magnetic Cataclysmic Variables

Since the discovery of X-ray emission from non-magnetic cataclysmic variables, improvements in observations have led to a greater understanding of the origin of X-rays emitted from these systems. Soft X-rays were originally detected from SS Cyg (the X-ray brightest dwarf nova) by Rappaport *et al.* (1974) during a rocket flight scan, and hard X-rays were then detected by Heise *et al.* (1978). Since then, samples of cataclysmic variables observed by various X-ray observatories have been published for *HEAO-1* (Cordova, Jensen & Nugent 1981), *Einstein* (Cordova, Mason & Nelson 1981), *ROSAT* (Van Teeseling & Verbunt 1994, Richman 1996, & Verbunt *et al.* 1997), and *EXOSAT* (Mukai & Shiokawa 1993), and I discuss the results of these surveys in the following paragraphs.

The accretion disk, which dominates the optical and ultraviolet emission from non-magnetic cataclysmic variables (figure 1.10), is expected to radiate half of the available gravitational potential energy (see section 1.5 for the mathematical proof). The remaining half of the gravitational potential energy is expected to be released through shocks in a narrow boundary layer, close to the surface of the white dwarf (see figure 1.13), heating the material up to temperatures of order  $10^5 K$  to  $10^8 K$  thus emitting X-rays.

These temperatures can be very approximately (yet simply) verified through considering the gravitational potential energy available to a particle falling towards the white dwarf from the inner Lagrangian point between the two stars. The available luminosity due to the gravitational potential is  $\dot{M}gh$ , where  $\dot{M}$  is the accretion rate,  $g$  is the acceleration due to gravity (where  $g = GM_{wd}/R_{wd}^2$ ,  $G$  is the gravita-

tional constant,  $M_{wd}$  &  $R_{wd}$  is the mass and radius of the white dwarf respectively), and  $h$  is the height the particle falls to the white dwarf (e.g. the distance to the inner Lagrangian point from the surface of the white dwarf). Approximating that the boundary layer emits radiation as a blackbody uniformly around the surface of the white dwarf, the Stefan-Boltzmann law can be used to estimate a temperature from this gravitational luminosity viz.  $\dot{M}gh = 4\pi R_{wd}^2 \sigma T^4$ . Rearranging for  $T^4$  gives an estimate of the temperature of  $\approx 10^8$  K (assuming  $\dot{M} = 10^{16} \text{ g s}^{-1}$ ,  $M_{wd} = 1.0$  solar mass, and  $R_{wd} = 10^4$  m).

This theoretical picture has passed some of the observational tests, with eclipse observations of the dwarf nova HT Cas showing that the size of the X-ray emitting region is comparable to, or smaller than, that of the white dwarf (Mukai *et al.* 1997). The density of this region (the boundary layer) is still unknown for either quiescence or outburst, although Mukai *et al.* (1997) calculate an lower limit of  $2.5 \times 10^{13} \text{ cm}^{-3}$  for the boundary layer of HT Cas (based upon the upper limit of the boundary layer size - the density could be much higher), and Done & Osborne (1997) point out that the models of Popham & Narayan (1995) suggest a density of  $\approx 10^{15} \text{ cm}^{-3}$  for SS Cyg in quiescence.

Early X-ray observations of cataclysmic variables only yielded primitive spectral information, although the sample of CVs observed with *HEAO-1* and *HEAO-2* (*Einstein*) did show that X-ray emission from dwarf novae exhibit a two-component variable spectrum throughout the outburst cycle; a hard 1-20 keV thermal bremsstrahlung component is observed throughout (with the higher temperatures during quiescence, Cordova, Jensen & Nugent 1981), and a much softer component is prominent only during outburst with  $kT < 50 \text{ eV}$  (Cordova *et al.* 1980).

The strong hard component observed in low accretion rate systems ( $\dot{M} < 10^{16} \text{ g s}^{-1}$ , e.g. during dwarf novae quiescence and in the VY Scl nova-likes when the disk is in a low state) is due to the gas in the boundary layer being optically thin and at a low density, and so cooling is inefficient (Patterson & Raymond 1985a) and the gas reaches high temperatures ( $\approx 10^8$  K), radiating hard X-rays via free-free emission (see figure 1.14).

In high accretion rate systems ( $\dot{M} > 10^{16} \text{ g s}^{-1}$ ), such as dwarf novae in outburst and the permanent outbursting UX UMa type nova-likes, the boundary layer between the inner accretion disk and white dwarf surface is permanently optically thick (and dense), allowing the material to cool efficiently and emitting predominantly soft X-ray emission with  $\approx 10^5 \text{ K}$  blackbody-like radiation (Pringle 1977, Patterson & Raymond 1985b). Thus the hard X-ray emission is suppressed during outburst and strong soft X-ray emission is observed instead.

Although the X-ray luminosity has been observed to reduce in the majority of outburst observations (such as SS Cyg, see figure 1.11), with X-ray count rates up to ten times greater during quiescence, there are a few exceptions. The *ASCA* observations of SS Cyg presented in this thesis (see section ??), and *RXTE* observations of U Gem (see figure 1.12), both show the system to be X-ray brighter in outburst. In the case of SS Cyg, the quiescent observation appears to be unusually X-ray faint, suggesting that accretion onto the white dwarf may have greatly reduced, or even ceased.

Spectral modelling of the hard X-ray emission detected in *EXOSAT* observations by Mukai & Shiokawa (1993) are indeed consistent with bremsstrahlung radiation being the source of the X-ray emission, with a single plasma temperature (2-20 keV) causing both continuum emission and a 6.7 keV iron emission line. However, *Ginga* observations suggest that the temperature distribution in this boundary layer may be more complicated than at first thought. Yoshida, Inoue & Osaki (1992) observed SS Cyg during quiescence, and found a thermal bremsstrahlung spectrum including an iron emission line. But the

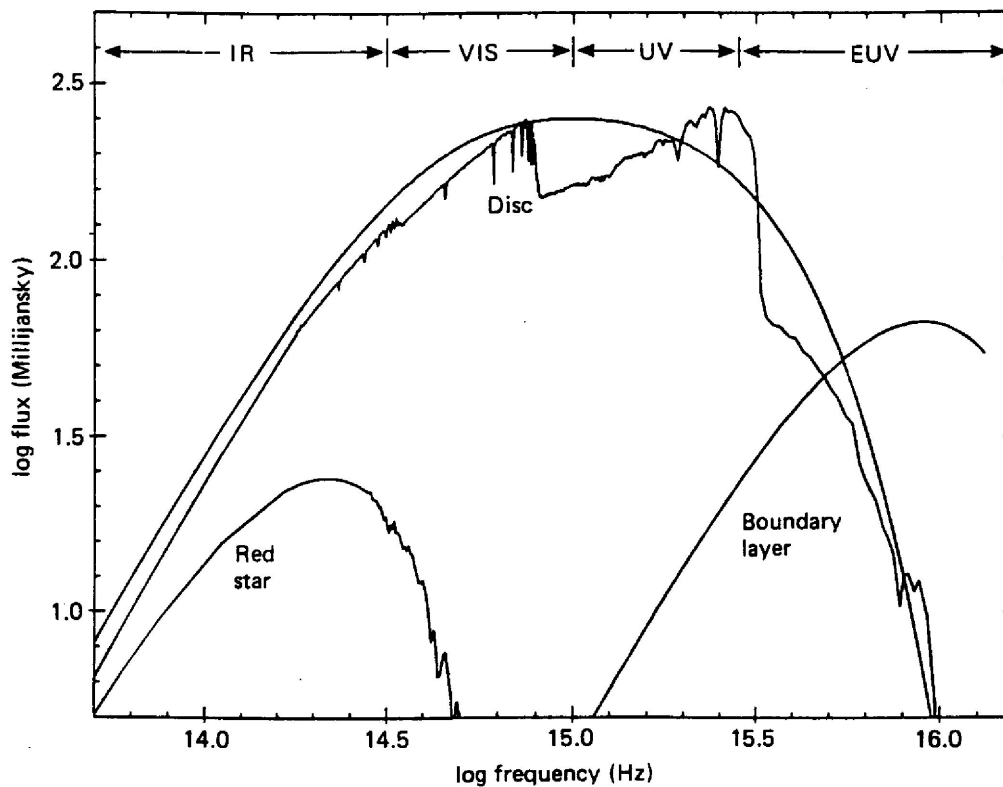


Figure 1.10: The broad band spectrum of non-magnetic cataclysmic variables (figure taken from Wade & Ward 1985). In order to visualised the graph in units more common to X-ray astronomy, the energy to frequency conversions are  $10^{14}$  Hz=0.41 eV,  $10^{16}$  Hz=41 eV, and the flux conversions are  $10^{1.0}$  mJy=1.51/E photons  $\text{cm}^{-2} \text{s}^{-1} \text{keV}^{-1}$ , and  $10^{2.5}$  mJy=477.5/E photons  $\text{cm}^{-2} \text{s}^{-1} \text{keV}^{-1}$ .

iron line central energy of SS Cygni appears to be inconsistent with the plasma temperature, and so Yoshida, Inoue & Osaki (1992) suggest that this hard component may be emitted by a two-temperature plasma at the source. The true temperature distribution of the X-ray emission from the boundary layer is a problem that is yet to be resolved.

A major problem with this theoretical picture of the X-ray emission originating from the boundary layer is that the observed X-ray luminosity is under-luminous when compared to the optical luminosity of the disk (see section ??). This directly contradicts the theoretical prediction that half the energy is radiated from the disk, and half from the boundary layer. Thus, the energy that is expected to be emitted as X-ray flux must be deposited elsewhere, either to drive an accretion disk wind or to spin up the white dwarf at the centre of the disc (this is discussed in more detail in chapter 6, section ??).

### 1.10.1 Absorption Features

Photoelectric absorption of X-rays along the line of sight (see figure 1.15) primarily effects lower energy X-ray photons, those below 2 keV. Many X-ray observations of cataclysmic variables reveal large amounts of absorption, much greater than that expected from interstellar gas alone. This may be due to a wind emitted from the surface of the accretion disk (see figure 1.16 and, for a review, Warner

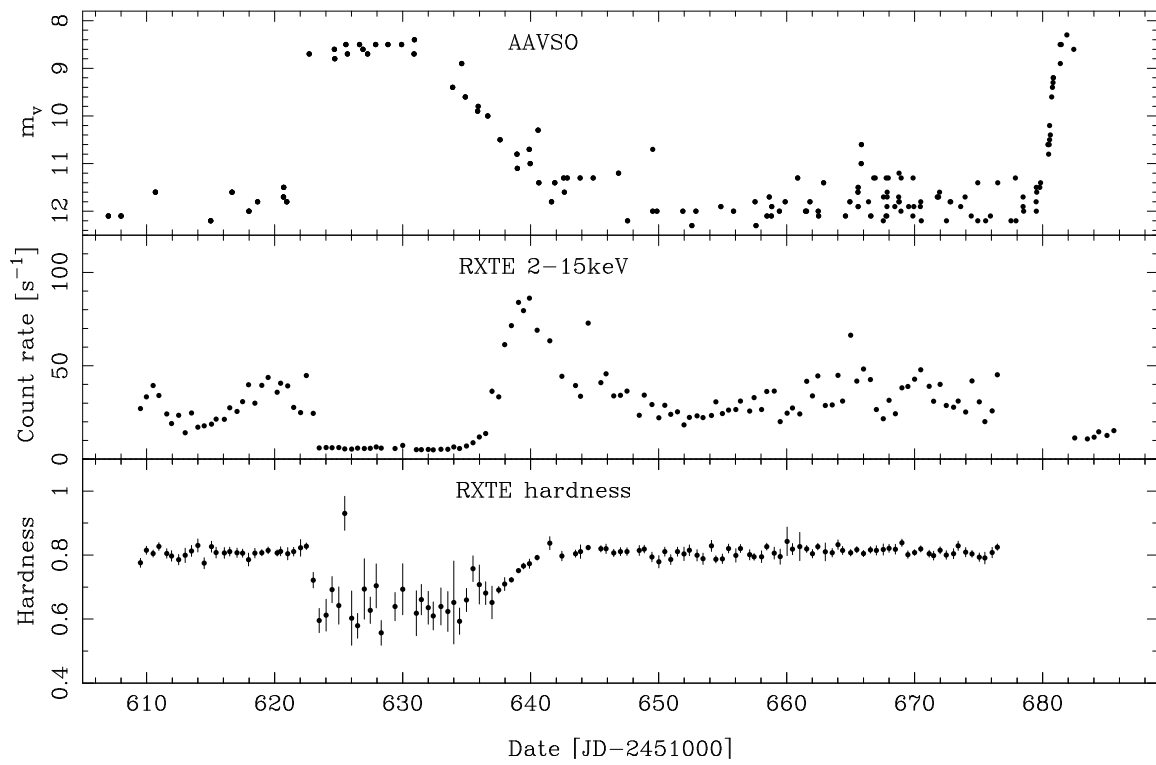


Figure 1.11: The X-ray light curve of SS Cyg during the *RXTE* monitoring campaign, showing the AAVSO optical data (top), the *RXTE* count rate (middle), and bottom, the *RXTE* X-ray hardness (Figure courtesy of P.J.Wheatley).

1995). *International Ultraviolet Explorer* observations of highly excited P Cygni profiles (spectral lines characterised by broad emission with a superimposed blue-shifted absorption trough, indicative of mass loss), predominantly in C IV and Si IV spectral lines, provide evidence of winds reaching speeds of  $\approx 5000 \text{ km s}^{-1}$  in many CVs. Such disk winds are expected to be observed when the accretion disk is in a high state (i.e. during dwarf novae outbursts or in permanent outburst nova-likes). Theoretical calculations (e.g. Proga, Stone & Drew 1999, see figure 1.16) suggest that accretion disk winds can reach large scale-heights above the accretion disk, and hence may be detectable as absorption in X-ray observations of cataclysmic variables even at quite low inclinations.

### 1.10.2 Reflections

Reflection signatures observed in the X-ray spectra of cataclysmic variables (e.g. in *ASCA* observations of SS Cyg, Done & Osborne 1997) are believed to be caused by X-rays reflecting (scattering) from the white dwarf surface or off the inner accretion disk. The simple picture used to describe the reflection physics is shown in figure 1.15. Such reflection is observed as a 6.4 keV fluorescent iron line (expected from photo-ionised neutral or mildly ionised iron up to Fe XV) and reflection of the continuum source spectrum. A reflection spectral model is shown in figure ??, and described in detail in section ??.

X-ray reflection signatures have been well documented in observations of active galactic nuclei (AGN) (Nandra & Pounds 1994). AGN comprise of two distinct sites for reflection; the inner geometrically

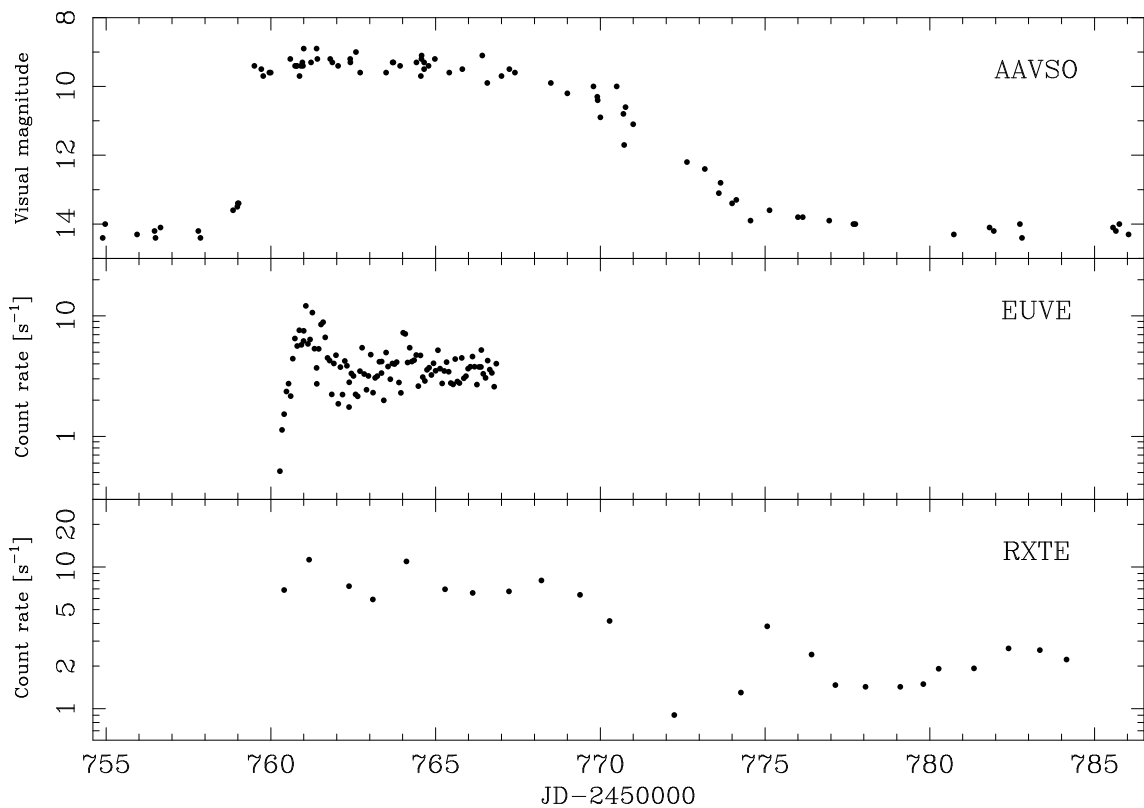


Figure 1.12: The AAVSO optical (top), *EUVE* (middle) and *RXTE* (bottom) light-curves of the dwarf nova U Gem during outburst (figure courtesy of Peter Wheatley). The hard (approximately 2-60 keV) X-ray count-rate was unusually brighter during the optical outburst than during the following optical quiescence period.

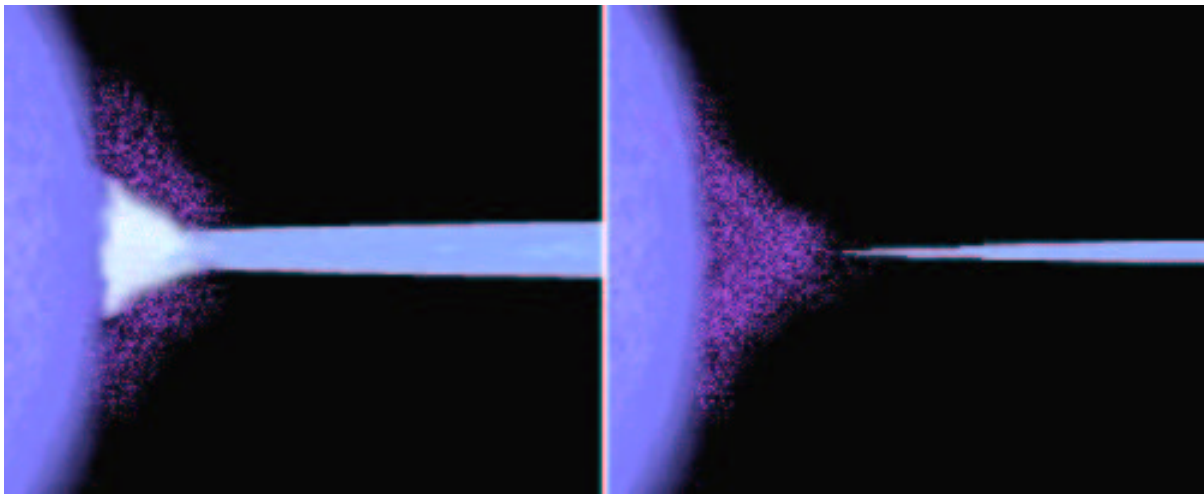


Figure 1.13: Origin of the X-ray emission in non-magnetic cataclysmic variables. In high accretion rate systems ( $\dot{M} > 10^{16} \text{ g s}^{-1}$ , pictured left) the boundary layer between the inner accretion disk and white dwarf surface is optically thick, emitting predominantly soft X-rays. In low accretion rate systems ( $\dot{M} < 10^{16} \text{ g s}^{-1}$ , pictured right), the density of the boundary layer remains low, the material does not cool efficiently, and hard X-rays are emitted.

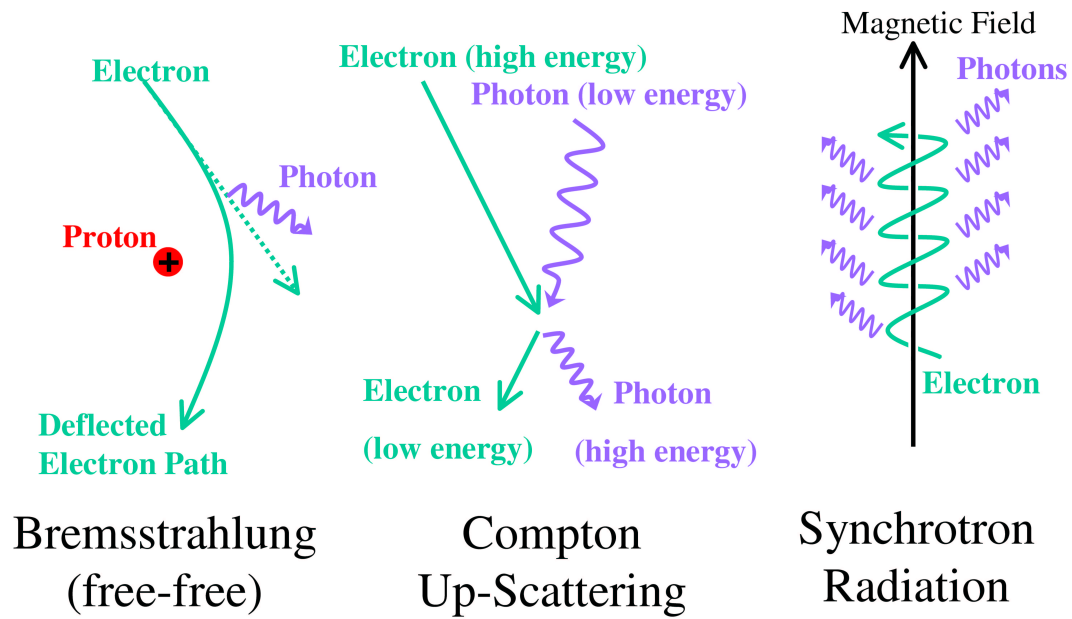


Figure 1.14: The astrophysical mechanisms for producing X-rays. Bremsstrahlung (free-free, or braking radiation) is the primary process for producing X-ray photons in non-magnetic cataclysmic variables. In disk-accreting galactic black-holes or supermassive black-hole (thought to be at the centre of active galactic nuclei), much harder energy X-rays can be produced via Compton scattering, where the kinetic energy of orbiting electrons is much greater than in cataclysmic variable systems. Astrophysical systems in the presence of strong magnetic fields (e.g. pulsars) can radiate X-rays via synchrotron emission.

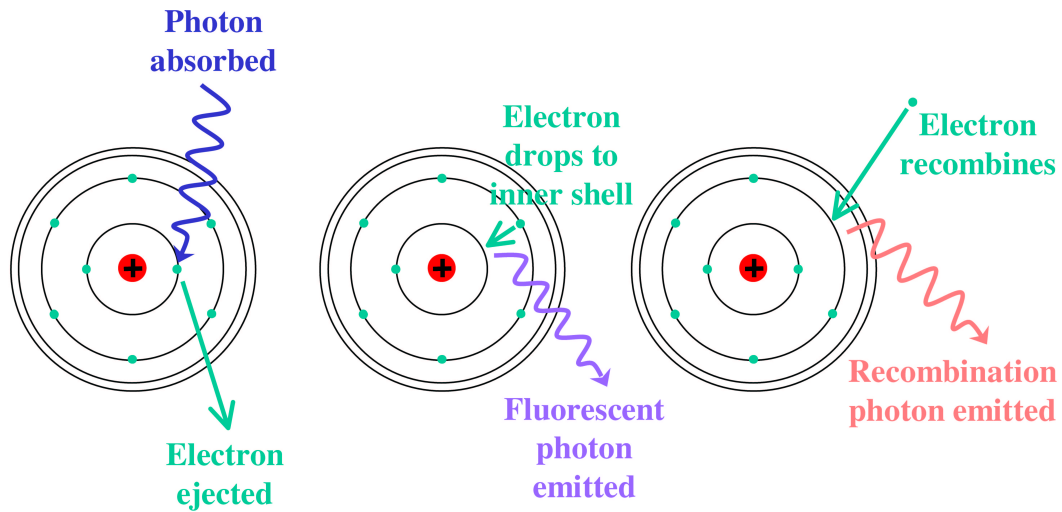


Figure 1.15: Atomic processes relevant to X-ray astronomy. Left: the atomic physics involved in X-ray photon absorption, predominantly by oxygen (around 1 keV) and iron (above 1 keV) atoms. Middle: when an electron is removed from an inner shell via irradiation (left), an outer electron then takes its place resulting in the emission of a fluorescent photon. Right: after an electron is ejected, the atom remains ionised until it recombines with an electron, producing line emission at an energy which is dependent on the ionisation state (and hence temperature) of the material.

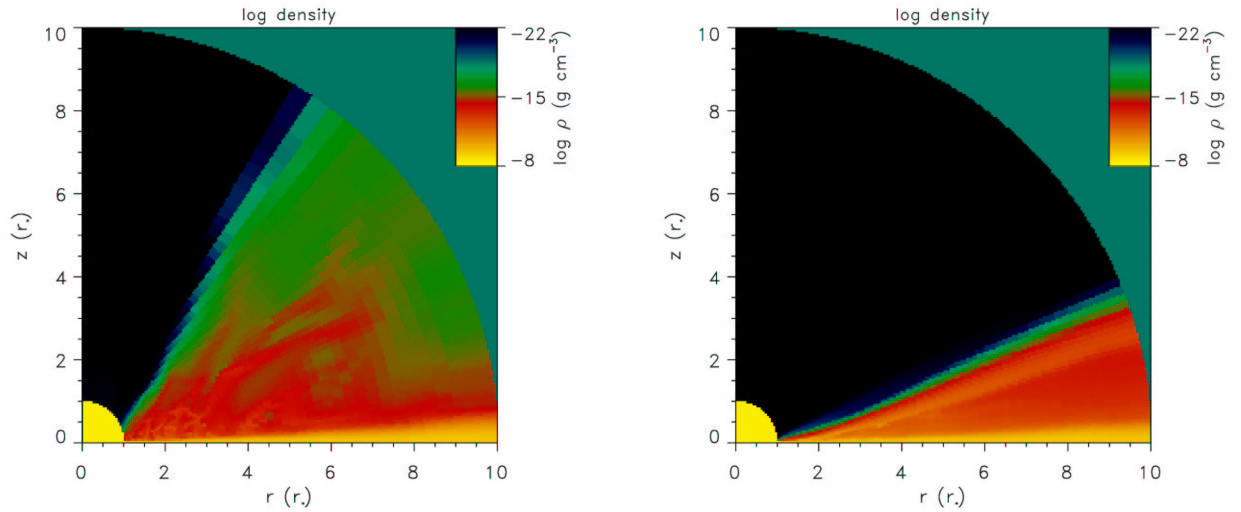


Figure 1.16: Simulations of accretion disk winds (figure taken from Proga, Stone & Drew 1999). In the unsteady solutions of the accretion disk models (left panel, see Proga, Stone & Drew 1999 for full details) the wind reaches large scale heights above the accretion disk. For steady accretion disk models (right panel, see Proga, Stone & Drew 1999 for full details), the disk-wind would not be detected in lower inclination cataclysmic variables.

thin accretion disk close to the black hole, and an outer geometrically thick accretion disk (the inner part of which is often described as a torus). The reflection off the inner disk is thought to occur off ionised gas, whereas the reflection occurs off neutral (or near neutral) gas in the torus (see figure 1.14). When X-rays impinge upon either the inner accretion disk or the outer "torus", Compton reflection modifies the spectrum via Compton scattering and bound-free absorption (Magdziarz & Zdziarski 1995). In Seyfert 2 galaxies, the continuum emission is suppressed due to the high levels of absorption in the geometrically thick "torus", allowing the source continuum that reflects off material situated away from the plane of the system to dominate the X-ray spectra.

When the reflection signature dominates the X-ray spectrum as it does Seyfert 2 galaxies, it allows accurate modelling of X-ray reflection signatures without the additional complication of modelling the continuum source. Hence reflection models developed for modelling AGN spectra are also useful when modelling the spectra of CVs (see Done & Osborne 1997).

However, the geometry of the reflection is somewhat different to that observed in cataclysmic variables. In CVs, the reflection is believed to occur off the white dwarf surface and/or inner accretion disk, and although the reflection in Seyfert 2's can occur off strongly ionised material where we would expect the reflecting material to be at least partially ionised, in heavily absorbed Seyfert galaxies scattering occurs off cold gas, well away from the X-ray source.



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